

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 7 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTECPC701 - INDUSTRY 4.0 AND AUTOMATION

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 Industry 4.0 mainly focuses on:
A) Mass production using steam engines
B) Manual labor in manufacturing
C) Automation and data exchange in manufacturing
D) Mechanical machining only
- 2 Which of the following is NOT a pillar of Industry 4.0?
A) Cloud Computing
B) Cybersecurity
C) Blockchain Mining
D) Big Data and Analytics
- 3 What does IIoT stand for?
A) Integrated Internet of Technologies
B) Industrial Internet of Things
C) Internal IoT Technologies
D) Internet Information of Things
- 4 Which layer in IIoT is responsible for collecting data?
A) Communication Layer
B) Processing Layer
C) Sensing Layer
D) Analytics Layer
- 5 Software Defined Networking (SDN) helps in:
A) Designing hardware
B) Improving data communication
C) Creating sensors
D) Building robots
- 6 What is the main benefit of Human-Robot Collaboration?
A) Slower production speed
B) Reduced safety
C) Improved productivity and flexibility
D) Replacing all human workers

- 7 A Digital Twin is best described as:
 A) A physical spare part
 B) A virtual replica of a physical system
 C) A 3D model only
 D) An automatic email system
- 8 SCADA stands for:
 A) System Control And Device Automation
 B) Supervisory Control And Data Acquisition
 C) Secure Control And Data Analyzer
 D) None of the above
- 9 What is the role of an HMI in SCADA systems?
 A) Collect data from sensors
 B) Execute control actions
 C) Provide user interface for monitoring
 D) Transmit power
- 10 RTUs are used in SCADA to:
 A) Design user interfaces
 B) Remotely monitor and control field devices
 C) Store historical data
 D) Control PLCs
- 11 Big Data is used in Industry 4.0 for:
 A) Painting machines
 B) Sending messages
 C) Analyzing large amounts of data
 D) Manual inspection
- 12 Analytics layer in IIoT helps in:
 A) Display only colors
 B) Play video
 C) Understand and analyze data
 D) Delete files

Q2. Solve the following.

- A) Explain the concept of Industry 4.0 and describe its main pillars. 6
- B) Describe the architecture and layers of the Industrial Internet of Things (IIoT). 6

Q3. Solve the following.

- A) What is Human-Robot Collaboration (HRC)? Explain its types and industrial applications? 6
- B) What is a Digital Twin? Explain its features, benefits, and applications? 6

Q4. Solve Any Two of the following.

- A) Explain the architecture and components of a SCADA system. Also state its advantages? 6
- B) Compare the four industrial revolutions? 6
- C) What are the advantages and challenges of using IoT in industrial environments? 6

Q5. Solve Any Two of the following.

- A) Discuss the recent technological advancements in industrial robotics? 6
- B) Write the advantages and limitations of industrial automation systems? 6
- C) Compare PLCs, RTUs, and DCS used in industrial automation? 6

Q6. Solve Any Two of the following.

- A) Explain the concept of a Smart Factory and its key characteristics? 6

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| B) | What is Software Defined Networking (SDN)? Explain its role in IIoT? | 6 |
| C) | What is Cloud Robotics? Explain its benefits in industrial applications? | 6 |

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